

Tic Tac Toe Web Game

Work Plan

Java Servlet · JSP · Apache Tomcat · NetBeans
10-Week Implementation Schedule

Phase Overview

Week	Phase	Weekly Goal
Week 1	Setup	Install tools, create NetBeans web project, push skeleton + README to GitHub
Week 2	Design	Sketch wireframe, draw UML class diagram, document MVC architecture in DESIGN.md
Week 3	Development	Build GameLogic.java with char[3][3] board, makeMove(), resetBoard(); register servlet in web.xml
Week 4	Development	Build clickable JSP board; implement GameServlet.doPost() with PRG redirect
Week 5	Development	Add checkWinner() (8 combos), isDraw(); display winner/draw banner; disable board on end
Week 6	Enhancement	Add New Game reset button, highlight winning cells in JSP, colour-code X and O in CSS
Week 7	Enhancement	Add session-based score tracker, live scoreboard label, dark-theme UI polish
Week 8	Testing	Write TEST_CASES.md, run all 8 test cases, fix bugs (one commit per fix), tag v1.0-tested
Week 9	Submission	Add Javadoc to all classes, write REPORT.md, take screenshots, finalise README
Week 10	Submission	Deploy final WAR, create GitHub Release with .war attached, submit package, prepare demo

Git Milestone

Git Tag	Week	What it represents
v0.1-skeleton	1	Empty WAR project pushed; Tomcat confirmed working
v0.2-board	4	Clickable JSP board; servlet receives moves; PRG redirect works

v0.3-logic	5	Win and draw detection complete; game-end banner shown
v0.4-features	7	Reset, highlight, score tracker, polished UI
v1.0-tested	8	All test cases pass; no known bugs
v1.0 (Release)	10	Final submission package: WAR + source + report

Github Commit Convention

Prefix	When to use it
init:	First push: project skeleton, README, .gitignore
feat:	New functionality added (board, servlet, win logic, reset, score)
fix:	Bug resolved — one commit per bug, describe the bug in the message
docs:	Documentation only: README updates, DESIGN.md, REPORT.md, Javadoc
style:	CSS / visual changes only — no logic change
refactor:	Code restructured without changing behaviour
release:	Final submission tag; attach WAR to GitHub Release

Key Technical Reminders

PRG Pattern (Post–Redirect–Get)

After every doPost() always call response.sendRedirect("index.jsp").
This prevents the browser from re-submitting the form on page reload.
Store all state in the HTTP session, not in instance variables on the servlet.

Session state

Store your GameLogic object: session.setAttribute("game", gameLogic)
Read it back: GameLogic gl = (GameLogic) session.getAttribute("game")
Create a new one if absent (first visit or after reset).

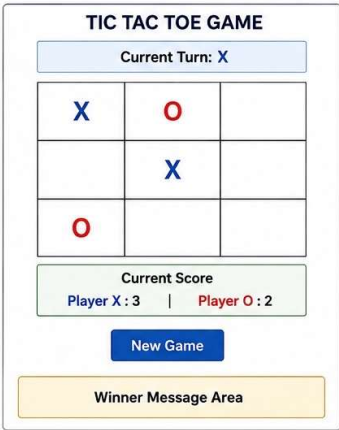
Win detection — 8 combinations to check

Rows : [0][0],[0][1],[0][2] | [1][0],[1][1],[1][2] | [2][0],[2][1],[2][2]
Cols : [0][0],[1][0],[2][0] | [0][1],[1][1],[2][1] | [0][2],[1][2],[2][2]
Diags : [0][0],[1][1],[2][2] | [0][2],[1][1],[2][0]

Diagram

WEEK 2: SYSTEM DESIGN AND ARCHIECTURE

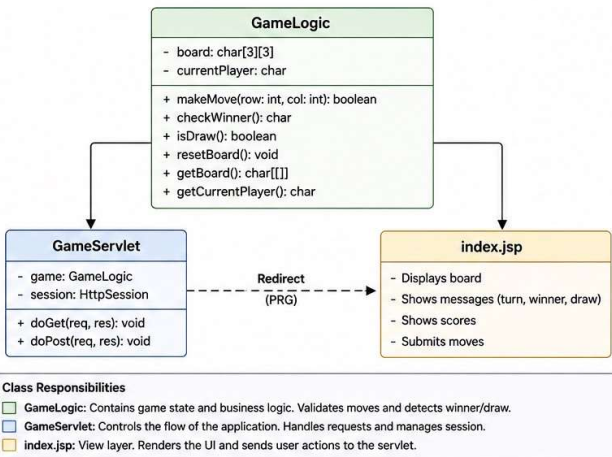
1. WIREFRAME DESIGN



Design Features

- Responsive 3x3 game board.
- Current player's turn displayed at the top.
- Winner or draw notification area.
- Session-based scoreboard.
- New Game button to reset the board.

2. UML CLASS DIAGRAM



3. MVC ARCHITECTURE (WITHOUT DATABASE)

